

ChromaIntel LC-MS/MS MVP Dashboard

Dataset size, model comparison, parameter significance, and development roadmap



Rows by source



Highest missingness fraction

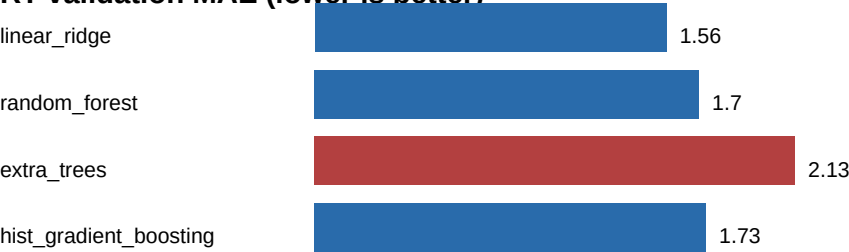


Executive Notes

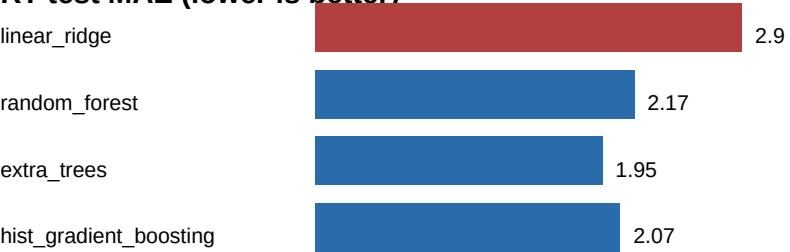
- Prediction targets: retention time in minutes (rt_min) and provisional peak quality score (quality_score).
- Current demo dataset is intentionally small and mixed-source; metrics are diagnostic, not production evidence.
- Internal bioanalytical runs with accepted and failed methods are the highest-value next data source.

Model Comparison

RT validation MAE (lower is better)



RT test MAE (lower is better)



RT model	Val MAE	Test MAE	Test R2
linear_ridge	1.558	2.901	-0.791
random_forest	1.703	2.168	-0.104
extra_trees	2.128	1.952	0.259
hist_gradient_boosting	1.733	2.072	-0.182

Quality model	Val MAE	Test MAE	Test R2
linear_ridge	0.034	0.069	0.386
random_forest	0.058	0.085	0.172
extra_trees	0.051	0.070	0.425
hist_gradient_boosting	0.068	0.099	-0.095

Source-wise RT error

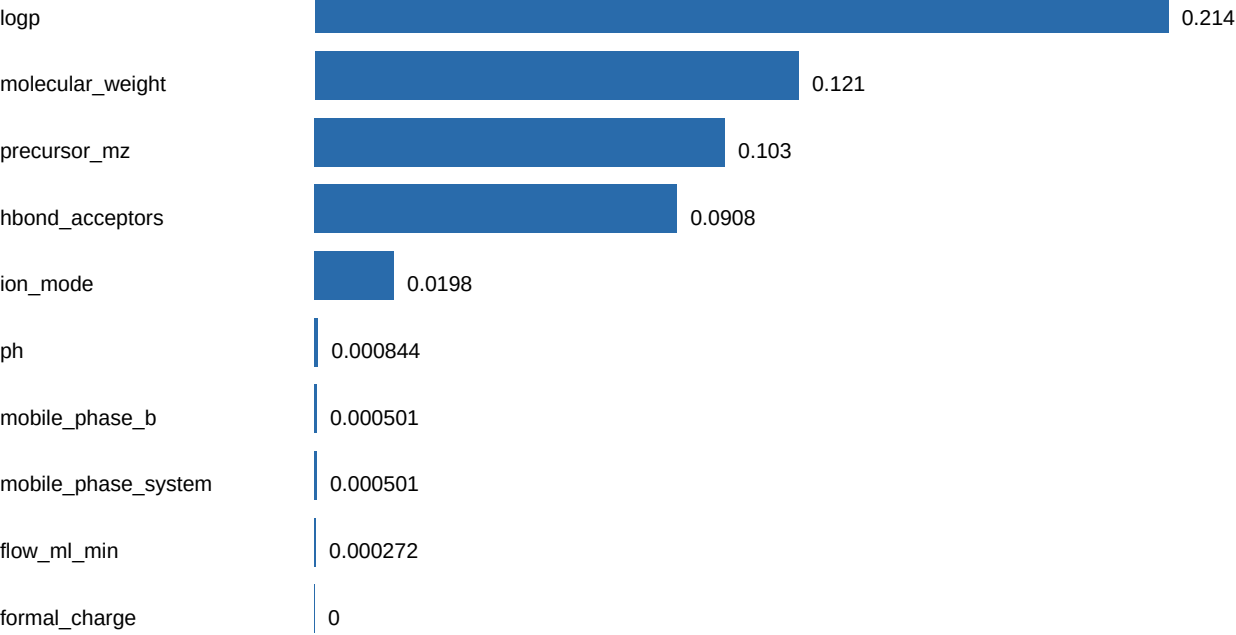
Source	N	MAE	Bias
METLIN_SMRT	1	5.006	-5.006
RepoRT	1	2.454	-2.454
internal_lab	1	1.242	1.242

Selected RT model: linear_ridge; selected quality model: linear_ridge

Uncertainty proxy: split_conformal_abs_residual_q90 q90=2.243 min

Parameter Significance And Validation Detail

Top RT permutation importance



Held-out prediction diagnostics

Test rows: 3 | RT MAE: 2.901 min | RT RMSE: 3.298 min | AD flags: 0

Compound	Source	Actual RT	Pred RT	Abs Error
Atorvastatin	METLIN_SMRT	7.250	2.244	5.006
Atenolol	RepoRT	3.200	0.746	2.454
Caffeine	internal_lab	1.350	2.592	1.242

Feature	Group	Imp.	Signal
logp	compound_descript	0.214	neutral_or_unsta
molecular_weight	compound_descript	0.121	neutral_or_unsta
precursor_mz	ms_setting	0.103	neutral_or_unsta
hbond_acceptors	compound_descript	0.091	weak_positive
ion_mode	ms_setting	0.020	neutral_or_unsta
ph	lc_numeric	0.001	neutral_or_unsta
mobile_phase_b	lc_categorical	0.001	neutral_or_unsta
mobile_phase_system	lc_categorical	0.001	neutral_or_unsta
flow_ml_min	lc_numeric	0.000	neutral_or_unsta
formal_charge	compound_descript	0.000	neutral_or_unsta

Roadmap

0-2 weeks: data foundation

- Import internal BE-style historical runs with accepted, failed, low-intensity, poor-resolution, and carryover cases.
- Expand reviewed public RT samples and preserve source/license/missingness provenance.
- Add import preview summaries for invalid rows, duplicate runs, and missing MS/LC fields.

2-4 weeks: stronger validation and models

- Move from random split to grouped/source holdout by InChIKey, source, and method family.
- Evaluate Morgan fingerprints, ExtraTrees, CatBoost/LightGBM/XGBoost when enough rows exist.
- Persist applicability-domain ranges and penalize out-of-domain recommendations.

1-2 months: recommendation quality

- Add score decomposition: RT fit, quality, runtime, confidence, AD penalty.
- Make LC search space configurable by lab-approved columns, solvents, pH, flow, temperature, and runtime.
- Collect outcome feedback from recommended methods for active learning.

2-4 months: SOTA research track

- Pretrain or benchmark graph/Transformer structure encoders on METLIN/RepoRT/MCMRT-scale RT data.
- Use retention-order objectives where absolute RT does not transfer cleanly between methods.
- Add calibrated conformal intervals by matrix, source, column chemistry, and instrument platform.

Decision gate: do not claim production readiness until internal data volume, grouped validation, and lab acceptance criteria are in place.